

Learned Timestep Sampling in Diffusion Models: Compute-Efficient Training via Adaptive Timestep Prioritization

Dhanvin S

Department of Computer Science
University of Maryland at College Park
College Park, USA
dhanvins@umd.edu

Basil Varghese

Department of Computer Science
University of Maryland at College Park
College Park, USA
basil106@umd.edu

Abstract—Diffusion models conventionally sample training timesteps uniformly, allocating equal compute regardless of difficulty. This paper proposes replacing the fixed uniform draw with a learnable sampling distribution that adaptively concentrates sampling on whichever timesteps remain hardest at each point during training. Because importance weighting is applied, the expected denoising objective remains equivalent to uniform training; the sampler instead alters sampling frequency and gradient variance across timesteps. We define parameterized softmax distributions $p(t; \theta)$ over $T=100$ timesteps, updated every 500 steps using loss-derived sampler objectives, with Lagrangian-style constraint penalties to prevent sampler over-concentration. A hybrid variant anchors 70% of sampling to uniform while allowing 30% to be learned. In a single-seed pilot study on CIFAR-10 with a UNet over 100,000 steps, hybrid learned sampling achieves FID 136.10 at just 10,000 steps, a quality level uniform training never reaches, while attaining the lowest FID AUC (182.13) of all three modes. However, the hybrid degrades in late training, with uniform yielding the best final-checkpoint FID (171.37). These results are suggestive of hybrid learned sampling as an early compute-efficiency technique, with open questions around late-training stability and generalization across seeds.

I. INTRODUCTION

Denoising diffusion probabilistic models (DDPMs) [1] learn to reverse a fixed noise process, training a neural network to predict noise added at each of T discrete timesteps. At each iteration the algorithm must decide: *which timestep t should we train on?* The standard answer is **uniform sampling**, drawing any $t \in \{0, \dots, T-1\}$ with equal probability $1/T$.

Uniform sampling is simple but wasteful. High-noise timesteps ($t \approx 80-99$) are difficult early but quickly mastered. Low-noise timesteps ($t \approx 0-10$) demand fine-grained pixel decisions and remain challenging throughout the run. Uniform allocation gives neither region adaptive attention.

This motivates a **learned sampler**: a parameterized distribution $p(t; \theta)$ that automatically concentrates sampling on the hardest timesteps at each training stage. Because importance weighting is applied (Section III-B), the denoiser’s expected objective remains identical to uniform training; the practical benefit is a change in gradient variance and optimization dynamics rather than a direct reweighting of expected loss.

Related ideas appear in loss-aware importance sampling [2] and curriculum learning [3], but fully learned constrained timestep samplers remain less explored in this setting.

We note an apparent tension in our results: the introduction observes that high-noise timesteps are quickly mastered early in training, yet the fully learned sampler concentrates on high-noise timesteps at the end. This is consistent because absolute MSE remains high at high-noise timesteps throughout training even after the denoiser improves, while perceptual quality (FID) is more sensitive to low-noise predictions. We return to this distinction in Section VI-D.

Our contributions are:

- **Softmax sampler**: $p(t; \theta)$ updated alongside the denoising network via separate loss-derived sampler objectives.
- **Constraint system**: Lagrangian-style penalties with adaptive penalty-multiplier updates (coverage + entropy floors) preventing sampler over-concentration.
- **Hybrid variant**: 70% uniform + 30% learned mixing for structural stability.
- **Empirical evaluation**: CIFAR-10 results showing strong early compute efficiency with a late-training tradeoff.

II. BACKGROUND

A. Diffusion Model Training

Starting from x_0 , the forward process produces noisy image x_t at timestep t :

$$x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I) \quad (1)$$

A neural network $\varepsilon_\phi(x_t, t)$ is trained to predict ε . Under uniform sampling:

$$\min_{\phi} \mathbb{E}_{t \sim U, x_0, \varepsilon} [\|\varepsilon_\phi(x_t, t) - \varepsilon\|^2] \quad (2)$$

B. Timestep Selection as Resource Allocation

Choosing t non-uniformly allocates compute across timesteps. Prior work explores fixed cosine schedules [4] and loss-weighted samplers [5], but fully learned constrained samplers remain less explored in this setting.

III. METHOD

A. Parameterized Sampler

We introduce $\theta \in \mathbb{R}^T$ with sampling distribution:

$$p(t; \theta) = \text{softmax}(\theta)_t = \frac{\exp(\theta_t)}{\sum_s \exp(\theta_s)} \quad (3)$$

Initializing $\theta = \mathbf{0}$ recovers uniform ($p = 1/T$). Any deviation is purely learned.

B. UNet Objective with Importance Weighting

Because $p(t; \theta)$ over-samples some timesteps, we correct via importance weighting:

$$\min_{\phi} \mathbb{E}_{t \sim p(t; \theta)} [w(t) \cdot \|\varepsilon_{\phi}(x_t, t) - \varepsilon\|^2] \quad (4)$$

where $w(t) = (1/T) / p(t; \theta)$. By construction, the importance-weighted expectation is identical to the uniform-sampling objective, so the denoiser's *expected* loss is unchanged. The practical effect of non-uniform sampling is that timesteps sampled more often appear in more minibatches, changing the variance and frequency of their stochastic gradient estimates even though the expected objective is preserved; this can accelerate optimization on persistently hard timesteps.

C. Sampler Objective

Every 500 steps we estimate per-timestep loss $\hat{L}(t)$ via EMA (exponential moving average). The target distribution is defined as the normalized square-root of per-timestep losses:

$$q_{\text{loss}}(t) = \frac{\hat{L}(t)^{0.5}}{\sum_s \hat{L}(s)^{0.5}} \quad (5)$$

The square-root dampens extreme concentration. The sampler minimizes cross-entropy toward this target:

$$\mathcal{L}_{\theta}(\theta) = \text{CE}(q_{\text{loss}}, p(\cdot; \theta)) + \text{constraint penalties} \quad (6)$$

Crucially, $\mathcal{L}_{\theta}$ is entirely separate from the UNet objective; θ is never updated by differentiating through the denoising loss.

Implementation note: The cross-entropy objective above (targeting q_{loss}) is the design used for the hybrid learned variant. The fully learned baseline uses an expected-loss objective that directly minimizes the per-timestep loss weighted by $p(t; \theta)$, rather than cross-entropy toward q_{loss} . This distinction means the two modes are not strictly comparable on the sampler objective alone; results should be interpreted as comparing the two overall training regimes rather than isolating the effect of the sampler objective.

D. Lagrangian-Style Constraint Enforcement

Left unconstrained, the sampler can over-concentrate probability on a small set of high-loss timesteps, reducing coverage of the full denoising chain. We enforce two conditions:

- **Coverage floor:** aggregate violation $V_{\text{cov}}(\theta) = \sum_t \max(0, \varepsilon - p(t; \theta))^2 = 0$
- **Entropy floor:** $H(p(\theta)) \geq H_{\min}$

where $H(p) = -\sum_t p(t) \ln p(t)$ is the Shannon entropy under natural logarithm, so the maximum entropy over $T = 100$

timesteps is $\ln(100) \approx 4.605$, and $H_{\min}$ is a tunable lower bound.

These are enforced via soft Lagrangian penalties with scalar multipliers $\lambda, \mu \geq 0$:

$$\mathcal{L}_{\theta}(\theta, \lambda, \mu) = \text{CE}(q_{\text{loss}}, p(\cdot; \theta)) + \lambda V_{\text{cov}}(\theta) - \mu (H(p(\theta)) - H_{\min}) \quad (7)$$

Multipliers λ and μ are updated using adaptive penalty-multiplier updates:

$$\lambda \leftarrow \max(0, \lambda + \alpha_{\lambda} V_{\text{cov}}(\theta)) \quad (8)$$

$$\mu \leftarrow \max(0, \mu + \alpha_{\mu} (H_{\min} - H(p(\theta)))) \quad (9)$$

Multipliers rise whenever a constraint is violated, exerting growing pressure until feasibility is restored.

E. Hybrid Variant

The fully learned sampler can still over-concentrate within the constraint budget. The hybrid mode adds two structural mechanisms:

Step 1 — Probability floor:

$$p_{\text{floor}}(t; \theta) = \varepsilon + (1 - T\varepsilon) \cdot \text{softmax}(\theta)_t, \quad \varepsilon = 0.002 \quad (10)$$

Step 2 — Uniform mixture:

$$p_{\text{final}}(t; \theta) = 0.7 \cdot \frac{1}{T} + 0.3 \cdot p_{\text{floor}}(t; \theta) \quad (11)$$

Note: $\varepsilon = 0.002$ is the floor on the *learned component*. The effective minimum of p_{final} is $0.7 \times 0.01 + 0.3 \times 0.002 = \mathbf{0.0076}$, the true per-timestep lower bound during hybrid training. The sampler objective in hybrid mode uses p_{final} and retains the adaptive penalty term $P(\theta)$:

$$\mathcal{L}_{\text{hybrid}}(\theta) = -\sum_t q_{\text{loss}}(t) \log p_{\text{final}}(t; \theta) + P(\theta) \quad (12)$$

where $P(\theta) = \lambda V_{\text{cov}}(\theta) - \mu (H(p(\theta)) - H_{\min})$ is the adaptive penalty evaluated on the learned distribution. Coverage is enforced both structurally through the probability floor and uniform mixture, and softly through $P(\theta)$ as an additional safeguard.

F. Update Schedule

Table I summarizes the update cadence. The UNet is updated every step to prevent stagnation; sampler and multipliers are updated every 500 steps to allow per-timestep loss estimates to stabilize.

TABLE I
UPDATE SCHEDULE

Variable	Method	Frequency
ϕ (UNet)	Adam	Every step
θ (sampler)	Adam	Every 500
λ, μ (multipliers)	Adaptive penalty-multiplier	Every 500

IV. EXPERIMENTAL SETUP

All experiments share one fixed configuration:

- **Dataset:** CIFAR-10 (32×32 , 10 classes)
- **Architecture:** UNet
- **Timesteps:** $T = 100$
- **Budget:** 100,000 gradient steps
- **Evaluation:** FID every 5,000 steps (20 checkpoints per mode)

Three modes differ only in how t is chosen each iteration:

- 1) **Uniform:** $p(t) = 1/100$ fixed; θ not updated.
- 2) **Learned:** $p(t; \theta) = \text{softmax}(\theta)_t$ with Lagrangian-style constraint penalties.
- 3) **Hybrid Learned:** p_{final} as in Section III-E, with retained soft penalties.

Metric. FID [6] measures generated vs. real distribution distance in Inception feature space. We also report *FID AUC* (area under the FID-vs-steps curve), where lower values indicate faster sustained convergence.

V. RESULTS

A. Quantitative Comparison

Table II summarizes performance across the three modes.

TABLE II
PERFORMANCE SUMMARY (100K STEPS, CIFAR-10)

Metric	Uniform	Learned	Hybrid
Best FID ↓	157.35	159.77	136.10
Step of best FID	20,000	45,000	10,000
Final FID (100k) ↓	171.37	176.22	196.60
FID AUC ↓	188.27	187.24	182.13
Best train loss ↓	0.0667	0.0431	0.0677
Final train loss ↓	0.0765	0.0620	0.0858



Fig. 1. FID vs. training steps. Hybrid (teal) reaches FID 136.10 at step 10,000 but degrades late. Uniform (grey) achieves the best final-checkpoint FID (171.37).

The hybrid mode achieves FID 136.10 at step 10,000, the best single checkpoint across the entire experiment and the only mode to cross FID 140. Its FID degrades in late training, finishing at 196.60 vs. uniform’s 171.37. The fully learned

mode tracks near-uniform in FID despite achieving the best training loss. All results are from a single training seed and should be treated as indicative rather than conclusive; FID estimates from 1,000 samples carry non-trivial variance.

B. Compute Efficiency

Table III shows the number of gradient steps required for each mode to first cross key FID thresholds.

TABLE III
STEPS TO FIRST REACH TARGET FID

Target FID	Uniform	Learned	Hybrid
≤ 175	20,000	15,000	10,000
≤ 160	20,000	45,000	10,000
≤ 150	Never	Never	10,000
≤ 140	Never	Never	10,000

Hybrid learned is the only mode to cross FID 150 or 140, and does so at just 10,000 steps, before either competing mode even reaches FID 175.

C. Learned Sampling Distributions

TABLE IV
FINAL DISTRIBUTIONS (STEP 100K)

Property	Uniform	Learned	Hybrid
$p(t)$ min–max	0.01–0.01	0.001–0.028	0.008–0.024
Entropy (max 4.605)	4.605	4.159	4.563
Top timesteps	All equal	$t = 98, 97, 94$	$t = 0, 1, 2$
Focus region	—	High noise	Low noise

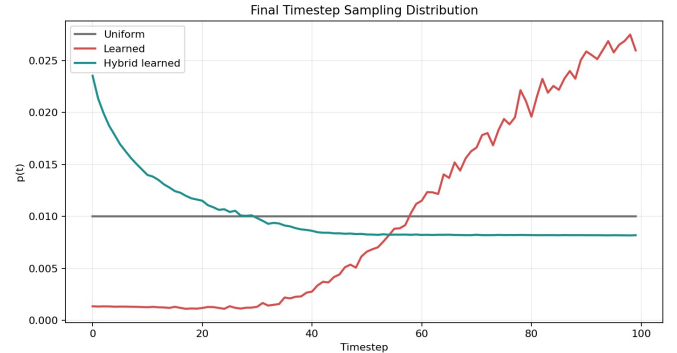


Fig. 2. Final $p(t; \theta)$ at step 100k. Learned (red) concentrates on high-noise end ($t \approx 60$ –99). Hybrid (teal) prioritizes low-noise end ($t \approx 0$ –10). Uniform (grey) stays flat at 0.010.

The two adaptive modes learned opposite strategies. The fully learned sampler concentrates on the noisiest end of the chain where absolute loss is largest; its entropy (4.159) drifts well below the uniform maximum. The hybrid, anchored structurally to uniform, redirects its learned 30% toward the persistently hard low-noise region driving the strong early FID gains while its entropy (4.563) stays near uniform (4.605).

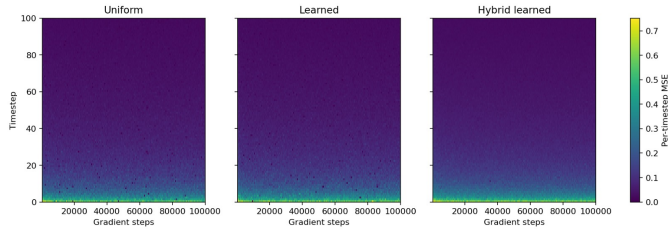


Fig. 3. Per-timestep MSE over training. Low-noise timesteps ($t \approx 0-10$) stay persistently hard across all three modes, motivating the hybrid’s low-noise focus.

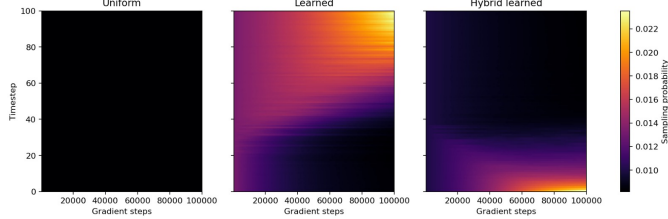


Fig. 4. Sampler evolution $p(t; \theta)$ over training. Uniform (left) stays constant. Learned (center) progressively concentrates on high-noise timesteps. Hybrid (right) maintains a steady low-noise emphasis.

VI. DISCUSSION

A. Early Efficiency vs. Late Stability

The core tradeoff is clear: adaptive sampling provides a strong early compute advantage but introduces instability in late training. The hybrid’s 70% uniform anchor mitigates but does not eliminate this. The fully learned mode avoids dramatic degradation but provides no FID advantage over uniform, as lower training loss does not translate to better sample quality here.

B. Why Opposite Focus Directions?

The fully learned sampler concentrates on high-noise timesteps in our run. Because this baseline uses an expected-loss sampler objective rather than the cross-entropy loss-target objective used by the hybrid mode, this behavior should be interpreted empirically rather than as a direct consequence of targeting high-loss timesteps.

The hybrid behaves differently: the 70% uniform anchor prevents the learned 30% from being dominated by the high-noise absolute loss signal. This is an empirical observation rather than a design guarantee — the uniform anchor appears to shift the sampler’s effective optimization landscape such that the residual learned component concentrates on the persistently hard low-noise region, where fine-grained prediction errors accumulate and FID sensitivity is highest. We hypothesize that removing the dominant high-absolute-loss signal from the learned component reveals the secondary structure of difficulty, but confirming this would require ablations across seeds and architectures.

C. Absolute MSE, Perceptual Quality, and FID

High-noise timestep performance (large absolute MSE, quickly improved relative to initialization) does not appear to be the binding constraint on FID for this model size and budget. Low-noise timesteps demand fine pixel-level decisions; errors there directly affect perceptual sharpness and structure. This distinction — between timesteps that are hard in absolute MSE vs. timesteps that are hard in their FID relevance — explains why the fully learned mode (high-noise focus) achieves the best training loss but not the best FID, while the hybrid (low-noise focus) achieves the best early FID despite a worse training loss.

D. Limitations

All experiments use a single random seed and a single architecture on CIFAR-10 at 100k steps. Results may not generalize to larger datasets, higher resolutions, or longer runs. FID is computed with 1,000 generated samples, so absolute values carry non-trivial variance; all comparisons are most meaningful as relative trends under the same evaluation protocol.

The fully learned and hybrid modes use different sampler objectives (expected-loss vs. cross-entropy toward q_{loss}), meaning they are not fully controlled comparisons of the sampler objective alone. Future work should align both modes on a single objective to isolate the effect of the uniform mixture.

The late-training degradation mechanism for the hybrid mode is not fully characterized. Possible causes include the learned component overfitting to transient loss spikes, the constraint multipliers drifting, or the 30% learned fraction being insufficient to maintain early gains as training progresses. Disentangling these would require per-component ablations.

Finally, the claim that non-uniform sampling improves compute efficiency rests on a change in gradient variance and update frequency rather than a change in expected loss. Whether this translates to wall-clock efficiency gains depends on implementation details not measured here.

VII. CONCLUSION

We introduced learned timestep samplers for diffusion model training, parameterized as softmax distributions and updated using loss-derived objectives, with Lagrangian-style constraint penalties preventing sampler over-concentration. A hybrid variant anchoring 70% of probability to uniform provides additional structural stability. Importance weighting ensures the denoiser’s expected objective remains equivalent to uniform training; the benefit of non-uniform sampling lies in altered gradient variance and update frequency across timesteps.

In a single-seed pilot study on CIFAR-10, results suggest that **hybrid learned sampling can improve early-training compute efficiency** substantially over uniform: FID 136 at 10,000 steps, a quality level uniform never reaches in 100,000 steps, and the lowest FID AUC of all modes. The claim of

superior final checkpoints does not hold; uniform wins end-of-run. These findings should be treated as suggestive given the single-seed setup and the objective mismatch between the learned and hybrid baselines.

Future work should explore: (1) aligning the sampler objective across all modes; (2) annealing the learned component toward uniform in late training; (3) a maximum-probability cap per timestep; (4) early-stop criteria tuned for adaptive runs; (5) multi-seed evaluation; and (6) scaling to larger architectures and datasets.

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